Exercise 1.1

1. Find $A \times B$, $A \times A$ and $B \times A$

   (i) $A = \{2, -2, 3\}$ and $B = \{1, -4\}$ (ii) $A = B = \{p, q\}$ (iii) $A = \{m, n\}$; $B = \phi$

2. Let A = {1,2,3} and B = $\{x \mid x$ is a prime number less than 10$\}$. Find $A \times B$ and $B \times A$.

3. If $B \times A = \{(-2,3),(-2,4),(0,3),(0,4),(3,3),(3,4)\}$ find A and B.

4. If $A = \{5,6\}$, $B = \{4,5,6\}$, $C = \{5,6,7\}$, Show that $A \times A = (B \times B) \cap (C \times C)$.

5. Given A={1,2,3}, B = {2,3,5}, C = {3,4} and D = {1,3,5}, check if $(A \cap C) \times (B \cap D) = (A \times B) \cap (C \times D)$ is true?

6. Let $A = \{x \in \mathbb{W} \mid x < 2\}$, $B = \{x \in \mathbb{N} \mid 1 < x \leq 4\}$ and $C = \{3,5\}$. Verify that

   (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

   (iii) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

7. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime number. Verify that

   (i) $(A \cap B) \times C = (A \times C) \cap (B \times C)$ (ii) $A \times (B - C) = (A \times B) - (A \times C)$

1.4 Relations

Many day-to-day occurrences involve two objects that are connected with each other by some rule of correspondence. We say that the two objects are related under the specified rule. How shall we represent it? Here are some examples,

Relationship	Expressing using the symbol R	Representation as ordered pair
New Delhi is the capital of India	New Delhi R India	(New Delhi, India)
Line AB is perpendicular to line XY	line AB R line XY	(line AB, line XY)
-1 is greater than -5	-1 R -5	$(-1, -5)$
ℓ is a line of symmetry for ΔPQR	ℓ R ΔPQR	(ℓ, ΔPQR)

How are New Delhi and India related? We may expect the response, "New Delhi is the capital of India". But there are several ways in which 'New Delhi' and 'India' are related. Here are some possible answers.

- New Delhi is the capital of India.
- New Delhi is in the northern part of India.
- New Delhi is one of the largest cities of India etc.,

So, when we wish to specify a particular relation, providing only one ordered pair

1 Relation and Functions.indd 6 15-12-2020 02:50:06 PM